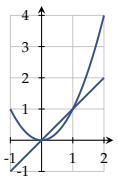
Sheet 24 Tue 15 Dec. 2026 *Solving $f(x) = g(x)$ and $f(x) < g(x)$*

1. Warm-up · Quick [EQ] Solve $2x - 1 = x + 3$.

   Ans. $x = 4$.

2. Core · Graph [REG] Curves of $f(x) = x$ and $g(x) = x^2$:

   Solve $f(x) = g(x)$ then $f(x) < g(x)$.

   Ans. $x = x^2 \Leftrightarrow x(x-1) = 0$: $x = 0$ or $x = 1$; $f(x) < g(x)$: the parabola is above the line for $x < 0$ or $x > 1$.

   Intervals: $]-\infty;0[\cup]1;+\infty[$.

3. Core + · Code reading [ALG] What does `f(-1) == g(-1)` return? And `f(0) == g(0)`?

   ```
   def f(x):
       return x*x
   def g(x):
       return 2*x + 3
   ```

   Ans. `True` ($1 = -2 + 3$) then `False` ($0 \neq 3$): -1 is a solution of $f(x) = g(x)$, 0 is not.

4. Review · Problem [MOD] Plan A: 15 000 Ar fixed + 1 000 Ar per session. Plan B: 2 500 Ar per session. From how many sessions is A cheaper?

   Ans. $15\,000 + 1\,000x < 2\,500x \Leftrightarrow x > 10$: from 11 sessions (10: equal at 25 000 Ar).

5. Thread · Quick [REP] Calculate the coordinates of the midpoint of $[AB]$ with $A(-2;5)$ and $B(6;1)$.

   Ans. $\left(\frac{-2+6}{2}; \frac{5+1}{2}\right) = (2;3)$.

One-variable descriptive statistics

Sheet 25 Thu 17 Dec. 2026 *Mean and linearity of the mean*

1. Warm-up · Quick [STA] Mean of 12, 15, 9, 14?

   Ans. $\frac{50}{4} = 12.5$.

2. Core · Method [STA] The series 4, 6, 8, 10, 12 has mean 8. What is the mean if 3 is added to each value? If each value is multiplied by 2?

   Ans. Linearity: $8 + 3 = 11$; $2 \times 8 = 16$.

   No need to recompute the sum.

3. Core + · True/false [STA] True or false? Justify. The mean of two numbers $a < b$ is always strictly between a and b.

   Ans. True. $a < \frac{a+b}{2} < b$ because $b - a > 0$.

4. Review · Reading [LEC] In a pie chart of 80 people, one sector represents 25 %. How many people does it represent? What is its angle?

   Ans. 20 people; 90°.

5. Thread · Quick [DIV] Calculate PGCD(84 ; 126) and deduce the simplest form of $\frac{84}{126}$.

   Ans. PGCD = 42; $\frac{84}{126} = \frac{2}{3}$.

Sheet 26 Mon 4 Jan. 2027 *Median and quartiles*

1. Warm-up · Quick [STA] Order 7, 2, 9, 4, 5, 11, 3 then give the median.

   Ans. 2 ; 3 ; 4 ; 5 ; 7 ; 9 ; 11; median 5 (rank 4).

2. Core · Method [STA] Series 3 ; 5 ; 6 ; 8 ; 9 ; 12 ; 14 ; 15. Give Q_1, the median, Q_3 and the interquartile range.

   Ans. $n = 8$: Q_1 has rank $\lceil 8/4 \rceil = 2$: 5; median $\dfrac{8+9}{2} = 8.5$; Q_3 has rank 6: 12; $Q_3 - Q_1 = 7$.
   Convention: the smallest value such that at least 25 % (resp. 75 %) of the data are less than or equal to it.

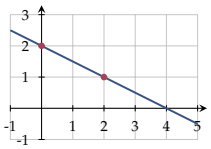
3. Core + · True/false [STA] True or false? Justify. At least half of the values of a series are greater than or equal to its median.

   Ans. True. By definition of the median (at least 50 % of the values on each side, endpoints included).

4. Review · Quick [CMP] Put 0.6, $\dfrac{5}{8}$ and 61 % in increasing order.

   Ans. $0.6 < 61\,\% = 0.61 < \dfrac{5}{8} = 0.625$.

5. Thread · Graph → formula [REG] The line below represents an affine function f. Determine $f(x)$.



   Ans. It passes through $(0;2)$ and $(2;1)$: slope $\dfrac{1-2}{2-0} = -\dfrac{1}{2}$, $f(x) = -\dfrac{1}{2}x + 2$.

Sheet 27 Tue 5 Jan. 2027 *Standard deviation; effect of adding or removing a value*

1. Warm-up · Quick [CAL] Calculate $(3-5)^2 + (7-5)^2$.

   Ans. $4 + 4 = 8$.

2. Core · Method [STA] Series 2 ; 4 ; 6: mean, variance and standard deviation (approximate value to 10^{-2}).

   Ans. Mean 4; variance $\dfrac{4+0+4}{3} = \dfrac{8}{3}$; standard deviation $\sqrt{\dfrac{8}{3}} \approx 1.63$.

3. Core + · True/false [STA] True or false? Justify. Adding 5 to each value of a series does not change its standard deviation.

   Ans. True. All the values and the mean are shifted by 5: the deviations from the mean are unchanged.

4. Review · Construction [CON] Let ABC be a triangle, I the midpoint of $[AB]$ and D the reflection of C in I. What type of quadrilateral is $ACBD$?

   Ans. A parallelogram: its diagonals $[AB]$ and $[CD]$ have the same midpoint I.

5. Thread · Quick [GEO] $(MN) \parallel (BC)$, $M \in [AB]$, $N \in [AC]$, $AM = 3$, $AB = 5$, $AC = 10$. Calculate AN.

   Ans. Thales: $\dfrac{AN}{AC} = \dfrac{AM}{AB}$, so $AN = 10 \times \dfrac{3}{5} = 6$.

Sheet 28 Thu 7 Jan. 2027 *Comparing series*

1. Warm-up · Quick [STA] Range of 3, 8, 5, 12?

   Ans. $12 - 3 = 9$.

2. Core · Interpret [STA] Two series A and B have the same mean 12; $\sigma_A = 1.5$ and $\sigma_B = 4$. Which one is more consistent?

   Ans. A: the values are closer together around the mean (smaller σ).

3. Core + · Interpret [STA] A: median 11, interquartile range 2. B: median 13, interquartile range 6. Compare the two series.

   Ans. B is centred higher (median $13 > 11$) but more spread out (interquartile range $6 > 2$).

4. Review · Quick [PRB] Two fair coins are tossed. What is the probability of getting two heads?